

Instructions:

1. Attempt the two sections in separate answer books.
2. Figure to the right indicate marks.
3. Make suitable assumptions wherever necessary and state them.

SECTION - I

Q.1 Do as directed.

1. Bookkeeping is depending on accountancy. **State True or False.** [01]
2. Balance sheet is a statement. **State True or False.** [01]
3. Minimum 2 members are requiring for partnership. **State True or False.** [01]
4. HUF stands for _____. [01]
5. Differentiate Sole trade and Partnership. [03]
6. Write down classified account for following given accounts. [04]
Building Account, Rent Account, Sina's Account, Vehicle's Account

Q.2 Answer the following questions.

[12]

- [A] List and explain characteristics of Partnership.
- [B] List and explain advantages and disadvantages of double entry system.

OR

Q.2 Answer the following questions.

[12]

The details related to Beta company are as below.

Selling price is Rs. 60 (per unit).

Variable cost is Rs. 20 (per unit).

Fixed cost is Rs. 400000.

Calculate followings.

- 1) Break Even Point in rupees.
- 2) Break Even Point in units.
- 3) Break Even Point in units, if variable cost is reduced by 50%.
- 4) Break Even Point in rupees and units, if fixed cost is increased by Rs. 50,000.
- 5) Number of units to be sold required to earn net profit of Rs.1,50,000.
- 6) Break Even Point in units, if sales price is reduced by 50%.

Q.3 Answer the following questions.

- [A] List and explain life cycle of accountancy in detail. [06]
- [B] Differentiate Private company and Public company. [06]

OR

Q.3 Answer the following question.**[12]**Trail Balance of Ashish Gandhi as on 31st March 2016. Prepare final accounts.

Name of Account	Debit	Name of Account	Credit
Purchases	2,00,000	Capital	3,00,000
Salaries	10,000	Sales	2,50,000
Rent	7500	Sundry creditors	1,05,000
Insurance premium	1500		
Drawings	50,000		
Machinery	1,40,000		
Cash at bank	22,500		
Computers	1,25,000		
Furniture	50,000		
Cash	10,000		
Opening Stock	26,000		
Sundry debtors	12,500		
	<u>6,55,000</u>		<u>6,55,000</u>

Adjustments:

1. Closing stock is Rs. 39,000.
2. Rent outstanding Rs. 1,000.
3. Provide interest on capital @ 10% and on Drawings @ 8%.

SECTION – II**Q.4 Do as directed.**

1. Define Statistics, Mean, Median, Mode and Range. **[05]**
2. Find Mean, Median, Mode and Range for the given data: **[04]**
13, 18, 13, 14, 13, 16, 14, 21, 13
3. Define Probability. What is the probability of getting a number 2 when rolling a die? **[02]**

Q.5 Answer the following questions.**[12]****[A]** Calculate Mean, Median and Mode for the following frequency distribution table.

Class Interval	150 - 154	155 - 159	160 - 164	165 - 169	170 - 174	175 - 179	180 - 184	185 - 189
Frequency	5	2	6	8	9	11	6	3

[B] Formulate the LPP (Linear Programming Problem)

A diet is to contain at least 4000 units of carbohydrates, 500 units of fat and 300 units of protein. Two foods A and B are available. Food A costs 2 dollars per unit and food B costs 4 dollars per unit. A unit of food A contains 10 units of carbohydrates, 20 units of fat and 15 units of protein. A unit of food B contains 25 units of carbohydrates, 10 units of fat and 20 units of protein. Formulate the problem as an LPP so as to find the minimum cost for a diet that consists of a mixture of these two foods and also meets the minimum requirements.

OR

Q.5 Answer the following questions.**[12]**

- [A]** Write the dual of the following Linear Programming Problem.
Also prove that dual of dual problem will give primal problem.

Minimize $4x_1 + 2x_2 - x_3$

subject to

$$x_1 + x_2 + 2x_3 \geq 3$$

$$2x_1 - 2x_2 + 4x_3 \leq 5$$

$$x_1, x_2, x_3 \geq 0.$$

- [B]** There are 4 blue marbles, 5 red marbles, 1 green marble, and 2 black marbles in a bag. If one marble is selected randomly then find probability for following events.

- Selected marble is of black color.
- Selected marble is of blue color.
- Selected marble is of black or blue color.
- Selected marble is of red or green color.
- Selected marble is not green color.
- Selected marble is not red color.

Q.6 Answer the following questions.(Attempt any two)**[12]**

- [A]** Find Optimal Solution of given LPP (Linear Programming Problem) using graphical method.

Maximize $Z = 50x + 18y$

subject to constraints:

$$2x + y \leq 100,$$

$$x + y \leq 80,$$

$$x \geq 0, y \geq 0$$

- [B]** There are 18 tickets marked with numbers 1 to 18. Find the probability of selecting a ticket having the following property:

- | | |
|-----------------|------------------------------|
| 1) even number | 4) number divisible by 3 |
| 2) odd number | 5) number divisible by 6 |
| 3) prime number | 6) number not divisible by 4 |

- [C]** Find an estimate of the mean, variance and standard deviation of the following data for the marks obtained in a test by 88 students.

Marks (x)	0-10	10-20	20-30	30-40	40-50
Frequency (f)	6	16	24	25	17